Aniket Thakur

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TECHNICAL SKILLS

Embedded: ESP32, Arduino, embedded C, bare-metal programming, I2C, UART, PID
Software: Python, C/C++, Java, TensorFlow, PyTorch, OpenCV, MediaPipe, NumPy
Tools: Autodesk Inventor, Fusion 360, Altium Designer, Git, Supabase, Vercel

EDUCATION

McMaster University | Engineering I, Bachelor of Engineering (Co-op) | Expected graduation: 2030

EXPERIENCES

Computer Vision Member - MAC RoboMaster Sep 2026 - Present

McMaster student design team developing an autonomous Sentry robot for RoboMaster competition.

- Contributing to the autonomous Sentry robot's Python vision stack on Orange Pi for object detection and target tracking.

Edge AI Member - AIAA McMaster FireSat Sep 2026 - Present

AIAA McMaster design team developing a wildfire-detection payload for a high-altitude balloon.

- Assessing the data and preprocessing requirements for applying the Sundance wildfire model to dual-LWIR/broadband thermal inputs on a planned 35 km balloon payload.
- Scoping payload software for thermal preprocessing, onboard inference, health monitoring, and downlink logic.

PROJECTS

Gesture-Controlled Robot repo

- **Fixed left/right gesture misclassification and cut training time to under 1 second** using MediaPipe hand landmarks and a compact MLP.
- **Built a live gesture-control pipeline** connecting camera input, landmark extraction, classification, and Wi-Fi motor commands to an ESP32 robot.

Competition Robotics - Maze & Sumo

- **Placed 1st in Maze and 2nd in Sumo at FEP** with Arduino-based autonomous robots.
- **Reduced corner wedging and improved maze recovery** by tuning wall-following behavior, drive response, and edge-detection thresholds.

PID Self-Balancing Robot repo

- **Built pitch estimation and live PID tuning** on ESP32 with a complementary filter and serial controls for on-bench tests.
- **Isolated residual instability** to gearbox backlash and motor mismatch, preparing the system for a matched-motor re-test.

Bare-Metal AVR Firmware

- **Wrote register-level embedded C** to configure a microcontroller output and turn on its built-in LED.
- **Built and programmed ELF and Intel HEX firmware images** through the board's serial bootloader.

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